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OPTIMIZATION

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About the Company

Velora is an e-mobility startup building a software-driven corporate transportation platform focused on optimizing employee commutes. As enterprises scale, employee transportation becomes increasingly complex due to dynamic demand patterns, varying employee schedules, diverse vehicle fleets, and rising operational costs. Conventional transportation systems rely heavily on static routes, manual planning, or vendor-driven scheduling. These approaches lead to inefficient fleet utilization, increased travel time, poor adaptability to real-time demand, and higher operational expenses. Velora addresses this challenge through a backend-first optimisation engine supported by a lightweight web interface for operational visibility and control.

Company's Objective

Velora's objective is to design a scalable corporate mobility system that intelligently assigns employees to vehicles and generates optimized pick-up and drop-off routes under real-world constraints. The system must deliver measurable improvements in cost efficiency and travel time while remaining flexible to employee preferences and operational limitations.

Problem Statement

Velora invites participating teams to design and implement an end-to-end corporate mobility optimisation system, consisting of a backend optimisation engine and a web-based frontend interface.

The objective of the system is to compute an optimal employee transportation plan for a given planning window using structured input data provided through Excel files. These inputs represent real-world enterprise commute scenarios, including employee ride requests and available vehicle fleets.



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Based on the provided inputs, the system must autonomously determine:

- The assignment of employees to available vehicles, respecting capacity, preference, and priority constraints
- The optimal pick-up and drop-off sequence for each vehicle in the fleet

The backend component of the system is responsible for all optimisation, routing, and decision-making logic. It must process the raw input data, apply constraints and policies, and generate an optimized routing plan that minimizes overall system cost and travel time.

The web-based frontend is mandatory and must serve as the primary interface for interaction and demonstration. It should allow users to upload or select input Excel files, trigger the optimisation process, and visualize the resulting routes and assignments on a real-world map. The frontend must also clearly present the quantitative outcomes of the optimisation, including cost and time improvements.

The system should be designed in a modular and extensible manner, such that changes in demand, supply, or constraints can be handled without manual intervention. While advanced features such as dynamic re-optimisation or learning-based enhancements are encouraged, they are not mandatory.

In addition to the required backend and web-based frontend, teams may optionally develop a mobile application to demonstrate extended usability or product thinking. Mobile application development is not required for a valid submission and will be evaluated for bonus points only.



Input Definition

1. Demand Side (Employee Requests)

Each employee ride request will include the following attributes:

- User identifier
- User priority / level
- Pick-up location
- Destination location
- Available time window for travel
- Vehicle preference (premium or normal)
- Sharing preference (single, double, or triple)

2. Supply Side (Vehicle Fleet)

Each vehicle entry will include the following attributes:

- Vehicle identifier
- Fuel type (petrol, diesel, or electric)
- Vehicle mode (2-wheeler, 4-wheeler, or van)
- Seating capacity
- Cost per kilometer
- Vehicle performance history:
 - Average mileage
 - Average speed
 - Vehicle age
- Current vehicle location
- Vehicle availability time

Constraints

The generated transportation plan must satisfy the following constraints:

- Vehicle capacity limits must not be exceeded
- Employee time windows must be respected
- Sharing and vehicle preferences must be honored
- Vehicles operate from their current locations
- Higher-priority employees have stricter tolerance for delays



Expected Output

The solution must produce clearly defined and verifiable outputs at both the backend and web frontend levels.

1. Backend Outputs

After processing the input data provided through Excel files, the backend optimisation engine must generate the following outputs for each test case:

- A vehicle-wise assignment of employees, indicating which users are assigned to which vehicle
 - An ordered pick-up and drop-off sequence for every vehicle
 - Estimated pick-up time and drop-off time for each employee
 - Total distance travelled by each vehicle
 - Aggregate system metrics, including:
 - Total travel time
 - Total operational cost
 - A baseline cost estimate for the same set of trips using standard market pricing (e.g., average prices from services such as Ola, Uber, or Rapido)
 - Net cost savings achieved through optimisation, compared to the baseline
- These outputs must be exposed in a structured form (e.g., internal data objects or JSON) to enable seamless consumption by the frontend.

2. Web Frontend Outputs (Mandatory)

The web-based frontend must visually demonstrate the optimisation results in a clear and intuitive manner. At a minimum, the frontend must include:

- An interface to upload or select the input Excel files
- A map-based visualization showing:
 - Initial state with only pick-up and drop-off locations plotted
 - Optimized routes overlaid after backend processing
 - Distinct visual representation for different vehicles and routes



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A results view displaying optimisation outcomes, including:

- Total cost after optimisation
- Estimated baseline cost without optimisation
- Absolute and percentage cost savings
- Total travel time before and after optimisation (if applicable)

The frontend should clearly communicate how the routing and assignment decisions reduce cost and travel time compared to non-optimised commute scenarios.

3. Presentation Requirements

For presentation and demonstration rounds, teams must be able to:

- Load one of the provided Excel-based test cases
- Trigger the optimisation process
- Visually demonstrate route planning on a real map
- Explain cost and time improvements using the displayed metrics

While the visual design of the frontend may be minimal, the correctness, clarity, and interpretability of the displayed results are critical.

The solution must produce clearly defined and verifiable outputs at both the backend and web frontend levels.

Evaluation Criteria

Submissions will be evaluated based on a clearly defined and weighted rubric, with a strong emphasis on backend optimisation logic and the quality of route planning

1. Backend Optimisation and Routing Logic (60%)

The primary evaluation focus is on the backend optimisation engine. Teams will be assessed on:

- Correctness and effectiveness of vehicle assignment and route planning logic
- Ability to minimise total cost and travel time under given constraints



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- Quality of route optimisation, including handling of time windows, priorities, and sharing preferences
- Clarity and soundness of the optimisation approach

While teams may leverage external services for basic map rendering or distance estimation, building the core optimisation and routing logic from scratch is strongly encouraged. Reliance on third-party APIs for end-to-end route optimisation may limit flexibility and will be viewed less favorably compared to custom-built optimisation approaches.

2. Web Frontend and Visualisation (20%)

Teams will be evaluated on how effectively the web-based frontend demonstrates the optimisation results.

This includes:

- Clear visualization of pick-up and drop-off points on a real-world map
- Visual representation of optimized routes and vehicle-wise assignments
- Ability to clearly show how routes change after optimisation
- Overall clarity and interpretability of the visual output

Visual design aesthetics are secondary to correctness and clarity of information presented.

3. Data Handling and Output Metrics (20%)

Teams will be evaluated on the correctness and completeness of data handling, including:

- Proper ingestion and processing of Excel-based input data
- Accuracy of computed metrics such as total cost, travel time, and distance
- Correct calculation and presentation of cost savings compared to baseline market pricing
- Consistency between backend outputs and frontend-displayed results



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4. Bonus: Mobile Application (Up to +10%)

Teams may earn up to an additional 10% bonus by developing a mobile application that complements the mandatory web-based interface.

The mobile application must integrate with the same backend system and demonstrate meaningful functionality. Building a mobile application alone, without a working backend and web frontend, will not be awarded any points.

Submission Requirements

Each team must submit a complete, runnable system consisting of:

- Backend optimisation engine source code
- Web-based frontend source code
- Clear instructions to run the system locally
- Sample input Excel files and corresponding outputs
- A brief explanation of the optimisation approach and key design decisions

The submitted system must be capable of ingesting Excel-based inputs, performing backend optimisation, and visualizing results through the web frontend.

Submissions without a working backend and web frontend will be considered incomplete and will not be evaluated.